

Cortex Central AI Orchestrator - Architecture Specification Document v0.1.0

1. Overview and Design Principles

The Cortex system coordinates multi-model AI inference, dynamic routing across quality tiers, and subsystem integrations. It utilizes a SQLite database in WAL (Write-Ahead Logging) mode for local operational logs and model catalog metadata.

2. Subsystem Integration Contracts

Hippocampus Subsystem: Dedicated memory service operating on port 8001. Stores structured memories and embeddings using PostgreSQL with pgvector extension for semantic recall and knowledge graph relationships.

plane-slim Subsystem: Lightweight issue and task management service running on port 8011. Accessible via REST API authenticated with X-API-Key headers.

3. Security Guardrails

All requests pass through the 2-Layer SecurityShield. Layer 1 executes deterministic regex checks for command patterns. Layer 2 evaluates prompt intent using the local Ollama model (qwen2.5vl:7b).